

MAKESH K

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AI Engineer | GenAI & RAG | Python, FastAPI, LangChain ♦ [LinkedIn](#) ♦ [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building Generative AI applications, RAG pipelines, and real-time Computer Vision systems. Experienced in Python, LangChain, FAISS, ChromaDB, FastAPI, OpenCV, TensorFlow Lite, and GPU-based image processing. Built AI-powered recruitment workflows, document retrieval systems, and real-time camera enhancement pipelines with REST API integration.

TECHNICAL SKILLS

Programming: Python, SQL **AI/LLM:** GenAI, LLMs, LangChain, OpenAI GPT, RAG, Embeddings, Prompt Engineering

Vector Search: FAISS, ChromaDB **Computer Vision:** OpenCV, MediaPipe, TensorFlow Lite, Image Processing, GPU Shaders

ML/DL: TensorFlow, Keras, scikit-learn, PyTorch, CNN, ANN, Classification, Regression, Clustering

Backend: FastAPI, REST APIs **Cloud/DB:** AWS, GCP, MySQL, MongoDB, Supabase **Tools:** Git, GitHub, VS Code, Jupyter

EXPERIENCE

Freelance AI & Software Developer Feb 2025 – Present

- Build AI applications across Generative AI, Computer Vision, image processing, and Machine Learning.
- Build real-time camera enhancement pipelines using Kotlin, Android Camera APIs, OpenCV, and GPU shaders for exposure, white balance, denoising, tone mapping, sharpening, and saturation.
- Design MediaPipe and TensorFlow Lite segmentation workflows for face, person, and scene analysis.
- Troubleshoot tensor shape, normalization, segmentation, output indexing, and real-time memory/performance issues.

Data Scientist Intern – Rubixe AI Aug 2024 – Jan 2025

Bengaluru

- Built ML pipelines covering preprocessing, feature engineering, training, evaluation, and deployment.
- Developed REST API-based model serving workflows for real-time prediction and application integration.
- Worked with SQL, AWS, and GCP for ML workflows, training, inference, and data processing.

Teaching Assistant – Data Science Team Nov 2021 – Jun 2023

RVS Educational Trust, Coimbatore

- Performed data preprocessing, analysis, and ML model development; applied Regression, Classification, Clustering, ANN, and CNN.

SELECTED AI PROJECTS

Thozhil Koodam AI – AI-Powered Recruitment Assistant

- Building an AI-powered mobile recruitment application for candidate assistance, job discovery, resume processing, and intelligent job matching using LLMs, embeddings, retrieval, and vector search.
- **Stack:** Flutter, FastAPI, Supabase, LLM, Embeddings, Vector Search.

AI Camera Enhancement Pipeline

- Developed a real-time camera enhancement pipeline using OpenCV, GPU processing, and shader-based image processing for exposure, white balance, denoising, tone mapping, sharpening, saturation, and HDR-style enhancement.

RAG-Based Document Search System

- Built a RAG pipeline for document retrieval and context-aware question answering using FAISS, ChromaDB, LangChain, embeddings, and FastAPI REST APIs.

EDUCATION

M.Sc. Computer Science – Specialization in Data Science Jun 2019 – Apr 2021

RVS College of Arts and Science

B.Sc. Information Technology Jun 2016 – Apr 2019

Kamalam College of Arts and Science

Diploma in Electrical & Electronics Engineering May 2014 – Apr 2016

Rudraveni Muthuswamy Polytechnic College

CERTIFICATIONS

Data Scientist Internship – Rubixe AI ♦ Generative AI Foundations – Microsoft ♦ Certified Data Scientist – NASSCOM

FutureSkills Prime ♦ Building LLM Applications using Prompt Engineering